

In-Class Exercise Sheet 3

The material underlying these exercises is mainly based on Justin Zobel's book on "Writing for Computer Science", the chapter on "Hypotheses, Questions, and Evidence".

1. **Properties of a good hypothesis.** Find the following words in the word puzzle.

PRECISE, LOOSE, SPECIFIC, CONTRADICTIONARY,
LIMITATIONS, UNAMBIGUOUS

Words are hidden →, ↓, and ↘. Which of these characterize a good research hypothesis?



Puzzle created using <https://www.superteacherworksheets.com>.

2. Which statements are consistent with Zobel's guidelines for writing good hypotheses? Mark True or False.
- (a) The more loose a concept is, the easier it is to validate experimentally.
☐ True ☐ False
 - (b) A hypothesis should be stated so precisely that two readers interpret it in the same way.
☐ True ☐ False
 - (c) A good hypothesis should make clear what is not being claimed.
☐ True ☐ False

3. **Case studies.** For each hypothesis below, state whether it violates Zobel’s guidelines. Justify your answer.
- (a) Hypothesis A: “Our system improves database performance.”
 - (b) Hypothesis B: “Our system reduces average query latency by at least 20% on TPC-C¹ workloads under high contention.”
 - (c) Hypothesis C: “Our system improves performance in most realistic scenarios.”
4. **Choosing between hypotheses.** Complete the following statement:
- “Where two hypotheses fit the observations equally well and one is clearly simpler than the other, the simpler should be chosen.
- This principle is known as _____.”
5. **Presenting experiments.** Consider the typical structure of an experiments section in a computer science paper. For each statement, mark True or False.
- (a) A clear experiments section often separates setup, results, and interpretation.
☐ True ☐ False
 - (b) The experimental setup should explain the datasets, baselines, and evaluation metrics.
☐ True ☐ False
 - (c) A figure should be accompanied by an explanation of what it shows and why it matters.
☐ True ☐ False
 - (d) It is important to state limitations clearly.
☐ True ☐ False
 - (e) The experiments section should only present positive outcomes.
☐ True ☐ False
6. **Case study.** In the following, we consider a computational experiment for a *campus coffee recommendation system*. The system suggests suitable coffee options to students based on their preferences, time of day, and current café availability.
- The goal is to evaluate whether the proposed method improves recommendation quality and reduces the time needed for users to find a suitable option, compared with baseline approaches such as popularity-based recommendations.
- For each sentence below, mark the most appropriate part of an experiments section.
- (a) “We evaluate our method on data collected from five campus cafés and compare it against two baselines.”
☐ Setup ☐ Result ☐ Discussion

¹TPC-C is a standard database benchmark for online transaction processing (OLTP) systems.

- (b) “Our method reduces the average search time by 23% compared with the popularity-based baseline.”
☐ Setup ☐ Result ☐ Discussion
- (c) “This suggests that personalization is especially useful when users have strong dietary preferences.”
☐ Setup ☐ Result ☐ Discussion
- (d) “We measure recommendation quality using click-through rate and normalized discounted cumulative gain.”
☐ Setup ☐ Result ☐ Discussion
- (e) “The gains are smaller late in the day, possibly because several cafés have fewer items available.”
☐ Setup ☐ Result ☐ Discussion
- (f) “Figure 3 shows that the proposed method remains faster than all baselines as the number of users increases.”
☐ Setup ☐ Result ☐ Discussion

7. **Levels of equivlence.** In comparing computational experiments, comparing results is not trivial. For comparing data or program outputs, we can distinguish:

- **Bitwise identity:** identical representation (e.g., byte-for-byte identical, same MD5 hash)
- **Structural equivalence:** same content, possibly differing in order or formatting

For comparing programs, we distinguish:

- **Functional equivalence:** same outputs for the same inputs
- **Behavioral equivalence:** same observable behavior under a given observation model (e.g., including I/O, timing, or side effects)

Note that there are further forms of equivalence that we do not discuss here.

8. For each of the following scenarios, indicate the *strongest level of equivalence (including bitwise identity)* that holds.

- (a) Two JPEG files have identical MD5 hashes.
☐ Bitwise identity ☐ Structural equivalence ☐ None of these options
- (b) Two JSON files contain the same properties, but in different order.
☐ Bitwise identity ☐ Structural equivalence ☐ None of these options
- (c) Two XML files contain the same entities, but in different order.
☐ Bitwise identity ☐ Structural equivalence ☐ None of these options

(d) A Java and a Python program produce identical outputs for all tested inputs.

- ☐ Functional equivalence ☐ Behavioral equivalence ☐ None of these options

9. **Case study.** Consider the functions $f(x) = x \cdot 2$ and $g(x) = x + x$.

Are f and g functionally equivalent? Explain.

10. **Case study.** Two implementations of an algorithm are evaluated in runtime experiments:

Run	Algorithm A (ms)	Algorithm B (ms)
1	100	90
2	105	200
3	98	95

(a) Can we conclude that B is faster? Why or why not?

(b) Suggest one better way to compare the runtimes.

11. **Comparing two methods.** The table below summarizes the efficiency of two methods for learning English. Make a comparison between the two methods. In your description, make sure you use:

- At least one irregular comparative form (e.g. worst, fewest).
- All the following words: less, fewer, more, much, many.
- At least two adverbs (e.g. efficiently, quickly, fluently).

Below is an example in which the first few rows of the table are compared.

“In both methods the length of the study was the same (i.e. four months), however the number of participants in Method B was higher than in Method A: 421 and 375, respectively. Not as many words were learned in Method B than in ...”

	METHOD A	METHOD B
No. participants in study	375	421
Length of study	4 months	4 months
No. words learned that can be used actively	500	456
No. words that can be understood	3,000	1,500
No. of tenses learned	5	8
Level of fluency achieved	mid	low
% errors made when speaking	35%	15%
Writing ability	low	good
% understood while listening to authentic radio news	10%	20%

Exercise from Adrian Wallwork's book "English for Academic Research: Writing Exercises".